

Effective Fourier certificates for the full (e=4) Polydegree column

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Abstract

Let $\mathcal{G}_d$ denote the polydegree stratum of plane polynomial automorphisms over $\mathbb{C}$. Lewis, Perry and Straub gave an explicit coefficient-polynomial criterion implying

$$\mathcal{G}_{(d+4)} \subseteq \overline{\mathcal{G}_{(d,5)}}$$

and verified it for $2 \leq d < 20$. We prove the containment for every $d \geq 2$. The proof makes a previously qualitative Fourier-limit argument effective. After writing $d = 4m + r$, one residue class has an exact solution at the origin. For the other three classes, exact rational coefficient bounds and complex interval Newton estimates certify all 14,985 cases $5 \leq m < 5000$. A uniform analytic envelope proves every $m \geq 5000$. The finite ledger was recomputed under normal and optimized Python, producing byte-identical 17 MB certificates, and a separate fail-closed audit checks coverage, ordering, hashes and explicit outward-rounded interval endpoints. The uniform certificate uses exact rational arithmetic together with 256-bit Arb evaluation only for cancellation-sensitive Fourier constants. This is a computer-assisted proof of one Polydegree-containment column; it is not a result about the Jacobian or Hessian conjectures.

1. Statement and relation to earlier work

For $e \geq 1$, write $\mathbf{x} = (x_1, \dots, x_e)$. If $\mathbf{a} \in \mathbb{Z}_{\geq 0}^e$, put

$$|\mathbf{a}| = \sum_{s=1}^e a_s, \quad \mathbf{a} \cdot \mathbf{N} = \sum_{s=1}^e s a_s.$$

Following Lewis–Perry–Straub [1], define

$$g_{n,e} = \frac{1}{n+1} \sum_{\mathbf{a} \cdot \mathbf{N} = n} (-1)^{|\mathbf{a}|} \binom{n+|\mathbf{a}|}{a_1, \dots, a_e, n} \mathbf{x}^{\mathbf{a}}. \quad (1.1)$$

Their Theorem 4 implies the following specialization criterion.

Published input 1.1 (Lewis–Perry–Straub)

If there is $P \in \mathbb{C}^e$ satisfying

$$g_{d,e}(P) = \cdots = g_{d+e-2,e}(P) = 0, \quad g_{d+e-1,e}(P) \neq 0, \quad a_{d,e}(P) \neq 0, \quad (1.2)$$

where $a_{d,e}$ is their determinant polynomial, then

$$\mathcal{G}_{(d+e)} \subseteq \overline{\mathcal{G}_{(d,e+1)}}. \quad (1.3)$$

They verified (1.2) for $e = 4$ and $2 \leq d < 20$ [1, Theorem 13(b)]. Perry’s earlier dissertation verifies a different, weaker intersection statement through $d = 45$ [2]; it must not be read as the containment in (1.3).

Our main result is as follows.

Theorem A

For every integer $d \geq 2$,

$$\boxed{\mathcal{G}_{(d+4)} \subseteq \overline{\mathcal{G}_{(d,5)}}} \quad (1.4)$$

The new range is $d \geq 20$. Its proof is computer-assisted only on the finite interval $20 \leq d < 20000$; an analytic certificate covers all larger degrees. The residue class $d \equiv 1 \pmod{4}$ also follows from Edo’s known divisibility theorem [4]; the new work is in the other three residue classes.

The literature audit accompanying this article did not locate an earlier all- d proof of (1.4), but that is an absence finding rather than a priority theorem. We therefore make no claim of being first until specialists have checked the result and its historical positioning.

2. Reduction to a smooth common zero

Lewis–Perry–Straub define polynomials $\alpha_{i,j,e}$ and

$$a_{d,e} = \det(\alpha_{i,j,e})_{0 \leq i \leq e-1, \, d-1 \leq j \leq d+e-2}.$$

Direct coefficient differentiation gives

$$\alpha_{i,j,e} = -\frac{\partial g_{j+1,e}}{\partial x_{i+1}}. \quad (2.1)$$

Let

$$D_{d,e} = \left(\frac{\partial g_{d+r,e}}{\partial x_s} \right)_{0 \leq r \leq e-1, \, 1 \leq s \leq e}.$$

Then $a_{d,e} = (-1)^e \det D_{d,e}$. Let J_s be the minor of the first $e-1$ rows obtained by deleting column s .

Proposition 2.1 (Jacobian–Euler identity)

In $\mathbb{Z}[x_1, \dots, x_e]$,

$$sx_s a_{d,e} = \sum_{r=0}^{e-1} (-1)^{e+r+s-1} (d+r) g_{d+r,e} M_{r,s}, \quad (2.2)$$

where $M_{r,s}$ deletes row r and column s from $D_{d,e}$. On the zero locus of the first $e-1$ rows,

$$sx_s a_{d,e} = (-1)^s (d+e-1) g_{d+e-1,e} J_s. \quad (2.3)$$

Proof The polynomial $g_{n,e}$ is weighted homogeneous of weight n for $\text{wt}(x_s) = s$, hence

$$\sum_{s=1}^e sx_s \frac{\partial g_{n,e}}{\partial x_s} = n g_{n,e}.$$

Replace column s of $D_{d,e}$ by the weighted sum of all columns. Multilinearity kills every summand except the original s -column. Expanding the resulting determinant along that column and using $a_{d,e} = (-1)^e \det D_{d,e}$ proves (2.2). Equation (2.3) follows after the first $e-1$ rows vanish. $\square$

For $e = 4$, we work on $x_4 = 1$ and use $s = 4$. It therefore suffices to find a common zero of $g_{d,4}, g_{d+1,4}, g_{d+2,4}$ at which their anchored 3×3 Jacobian and $g_{d+3,4}$ are nonzero.

3. Exact rescaling

Write

$$d = 4m + r, \quad 0 \leq r \leq 3.$$

For each $n \in \{d, d+1, d+2, d+3\}$, write $n = 4M + q$, where $M \in \{m, m+1\}$ and $0 \leq q \leq 3$. Put

$$C_{M,q} = \frac{(-1)^M}{4M+q+1} \frac{(5M+q)!}{(4M+q)!M!} \quad (3.1)$$

and define the row used by the verifier,

$$P_{M,q}^{(m)}(X) = \frac{g_{4M+q,4}(X_1/m, X_2/m, X_3/m, 1)}{C_{M,q}}. \quad (3.2)$$

For $\mathbf{a} = (a_1, a_2, a_3)$, set

$$u = a_1 + a_2 + a_3, \quad w = a_1 + 2a_2 + 3a_3, \quad t = \frac{w-q}{4}, \quad \ell = u - t.$$

A term occurs precisely when $w \equiv q \pmod{4}$ and $0 \leq t \leq M$. Its coefficient before $X_1^{a_1} X_2^{a_2} X_3^{a_3} / (a_1! a_2! a_3!)$ is exactly

$$c_{M,q}^{(m)}(\mathbf{a}) = \frac{(-1)^\ell}{m^u} \prod_{i=1}^{\ell} (5M+q+i) \prod_{i=0}^{t-1} (M-i). \quad (3.3)$$

Choose

$$\rho = 5^{1/4} e^{\pi i/4}, \quad \rho^4 = -5.$$

Since

$$4\ell - q = 3a_1 + 2a_2 + a_3,$$

the limiting coefficient is

$$c_q^\infty(\mathbf{a}) = (-5)^\ell = \rho^q \rho^{3a_1+2a_2+a_3}. \quad (3.4)$$

The exact finite-to-limit ratio is

$$R_{M,q}^{(m)}(\mathbf{a}) = \prod_{i=1}^{\ell} \left(1 + \frac{q+i}{5M}\right) \prod_{i=0}^{t-1} \left(1 - \frac{i}{M}\right) \left(\frac{M}{m}\right)^u. \quad (3.5)$$

All analytic and finite bounds below start from (3.3) or (3.5); no fitted coefficient model is used.

4. Fourier limit and simple zeros

Define

$$H_q(X) = \frac{1}{4} \sum_{j=0}^3 i^{-qj} \exp\left(\rho^3 i^j X_1 + \rho^2 i^{2j} X_2 + \rho i^{3j} X_3\right), \quad (4.1)$$

and $L_q = \rho^q H_q$. The root-of-unity filter shows that L_q has the coefficients (3.4), so the four finite rows converge to the corresponding L_q .

Put

$$E_j = \exp\left(\rho^3 i^j X_1 + \rho^2 i^{2j} X_2 + \rho i^{3j} X_3\right).$$

For residue r , choose C_r with $C_r^4 = (-1)^{r-1}$ and impose

$$E_j = C_r i^{(r-1)j}. \quad (4.2)$$

The compatibility equation is $\prod_j E_j = 1$. Choose logarithms $\Lambda_{j,r}$ summing to zero and set

$$X_{s,r}^* = \frac{1}{4\rho^{4-s}} \sum_{j=0}^3 \Lambda_{j,r} i^{-sj}, \quad s = 1, 2, 3. \quad (4.3)$$

We use the following exact phase choices, specified by their logarithms divided by πi :

r	$(\Lambda_{0,r}, \dots, \Lambda_{3,r})/(\pi i)$
0	(1/4, -1/4, -3/4, 3/4)
2	(1/4, 3/4, -3/4, -1/4)
3	(-1/2, 1/2, -1/2, 1/2)

The residue $r = 1$ is treated exactly in Section 7. Fourier inversion gives

$$H_r(X_r^*) = H_{r+1}(X_r^*) = H_{r+2}(X_r^*) = 0, \quad H_{r-1}(X_r^*) = C_r \neq 0. \quad (4.4)$$

Moreover,

$$\frac{\partial H_q}{\partial X_s} = \rho^{4-s} H_{q-s}. \quad (4.5)$$

Thus the first-three-row limit Jacobian is diagonal after accounting for the row normalizations. The diagonal moduli and the fourth-row modulus are:

r	first-three-row diagonal moduli	fourth-row modulus
0	$(\rho ^3, \rho ^3, \rho ^3)$	$ \rho ^3$
2	$(\rho ^5, \rho ^5, \rho)$	$ \rho $
3	$(\rho ^6, \rho ^2, \rho ^2)$	$ \rho ^2$

Consequently the inverse row-sum norm is at most

$$|\rho|^{-1} < \frac{67}{100}. \quad (4.6)$$

The fourth-row limit modulus is at least

$$|\rho| > \frac{149}{100}. \quad (4.7)$$

A 256-bit Arb evaluation proves

$$|X_{1,r}^*| \leq \frac{1}{3}, \quad |X_{2,r}^*| \leq \frac{71}{100}, \quad |X_{3,r}^*| \leq \frac{3}{4} \quad (4.8)$$

for $r = 0, 2, 3$. These enclosures are reproduced by the uniform verifier.

5. Uniform effectivity for $m \geq 5000$

This section proves an explicit analytic handoff. Let

$$\Delta_{M,q}^{(m)} = P_{M,q}^{(m)} - L_q.$$

We bound Δ on the closed polydisc

$$\|X - X_r^*\|_\infty \leq \frac{1}{4}. \quad (5.1)$$

By (4.8), its coordinate moduli are at most

$$\left(\frac{7}{12}, \frac{24}{25}, 1 \right). \quad (5.2)$$

5.1 Head expansion through degree 50

Let $\omega = M - m \in \{0, 1\}$. Expanding (3.5) gives

$$R_{M,q}^{(m)}(\mathbf{a}) = 1 + \frac{A_{q,\omega}(\mathbf{a})}{m} + O(m^{-2}), \quad (5.3)$$

where

$$A_{q,\omega}(\mathbf{a}) = \frac{q\ell}{5} + \frac{\ell(\ell+1)}{10} - \frac{t(t-1)}{2} + \omega u. \quad (5.4)$$

The corresponding exponential differential operator can be evaluated by the four Fourier modes. More explicitly, for mode j , put

$$z_s = \rho^{4-s} i^{sj} X_s,$$

$$p_t = -\frac{q}{4} + \sum_{s=1}^3 \frac{s}{4} z_s, \quad d_t = \sum_{s=1}^3 \frac{s^2}{16} z_s,$$

$$p_\ell = \frac{q}{4} + \sum_{s=1}^3 \frac{4-s}{4} z_s, \quad d_\ell = \sum_{s=1}^3 \frac{(4-s)^2}{16} z_s, \quad p_u = \sum_{s=1}^3 z_s.$$

The multiplier corresponding to (5.4) is

$$\mathcal{A}_{q,\omega,j} = \frac{q}{5} p_\ell + \frac{p_\ell^2 + d_\ell + p_\ell}{10} - \frac{p_t^2 + d_t - p_t}{2} + \omega p_u, \quad (5.4a)$$

and therefore

$$G_{q,\omega}(X) = \frac{\rho^q}{4} \sum_{j=0}^3 i^{-qj} \mathcal{A}_{q,\omega,j} E_j. \quad (5.4b)$$

At each X_r^* , 256-bit Arb arithmetic proves that the largest first-order row defect is

$$1.884955592153876 \dots < 2. \quad (5.5)$$

On (5.1), an exact rational triangle calculation gives

$$\|G\|_\infty \leq \frac{51944278023}{204800000} = 253.6341700 \dots < 254. \quad (5.6)$$

For the second-order remainder, let

$$A^+ = \frac{q\ell}{5} + \frac{\ell(\ell+1)}{10} + \frac{t(t-1)}{2} + \omega u.$$

The elementary product estimate

$$\left| \prod_j (1 + \delta_j) - 1 - \sum_j \delta_j \right| \leq \frac{1}{2} \left(\sum_j |\delta_j| \right)^2 \exp \left(\sum_j |\delta_j| \right) \quad (5.7)$$

together with the difference between $1/M$ and $1/m$ gives the exact coefficient remainder

$$\left| R_{M,q}^{(m)} - 1 - \frac{A_{q,\omega}}{m} \right| \leq \frac{\frac{6}{5}(A^+)^2/2 + A^+}{m^2} \quad (5.8)$$

for $u \leq 50$ and $m \geq 5000$. Summing (5.8) exactly over all admissible head monomials gives

$$K_{2,\text{ball}} < 15374, \quad K_{2,\text{centre}} < 1105. \quad (5.9)$$

The verifier checks the stronger exact rational values $15373.3723513\dots$ and $1104.3997874\dots$

5.2 Tail above degree 50

On (5.1), the absolute limit level of total degree u is bounded by

$$B_u = \frac{15^u}{u!}. \quad (5.10)$$

Indeed, after dropping the congruence restriction, the multinomial theorem bounds the level by

$$\frac{|\rho|^q}{u!} \left(|\rho|^3 |X_1| + |\rho|^2 |X_2| + |\rho| |X_3| \right)^u.$$

The bounds $|\rho| < 3/2$ and (5.2) make this smaller than (5.10) for $u \geq 51$. For a finite coefficient, the falling-factor product in (3.5) is at most one, while

$$\prod_{i=1}^{\ell} \left(1 + \frac{q+i}{5M} \right) \left(\frac{M}{m} \right)^u \leq \exp \left(\frac{(u+3)^2}{2m} \right).$$

This proves the finite majorant used below.

For $u \geq 51$, the first-order multiplier in (5.4) is at most u^2 . Indeed,

$$A^+ \leq \frac{61}{160}u^2 + \frac{17}{10}u \leq u^2. \quad (5.11)$$

Here we used $q \leq 3$, $\ell \leq u$, $t \leq 3u/4$ and $\omega \leq 1$. The last inequality already holds for $u \geq 3$.

For $u \leq m$, the finite level is bounded by

$$B_u \exp \left(\frac{(u+3)^2}{2m} \right). \quad (5.12)$$

The verifier sums the finite, limit and m^{-1} -scaled first-order levels exactly for $51 \leq u \leq 400$. Starting at $u = 401$, the limit and first-order successive ratios are bounded by $1/20$; the same is true for the finite majorant while $u \leq m$. Its successive ratio is

$$\frac{15}{u+1} \exp \left(\frac{2u+7}{2m} \right),$$

which decreases on this range. The three first-ratio inequalities are checked in exact rational arithmetic, with a rational Taylor upper bound for the one exponential. For $u > m$, use $u! \geq (u/e)^u$, $M \leq m+1$ and $q \leq 3$. On the polydisc (5.1), put $R_X = \max_s |X_s|$; (5.2) gives $R_X \leq 1$, hence

$3R_X \leq 3$. For every $m \geq 5000$ and $u \geq 5001$, the direct finite level is below $(13/1000)^u$: using $e < 3$, its level base is strictly smaller than

$$\frac{45}{u} + \frac{9}{m} + \frac{72}{mu} < \frac{13}{1000}.$$

Consequently the complete omitted contribution is

$$T = 2.482176974351622 \dots \times 10^{-6} < \frac{1}{400000}. \quad (5.13)$$

The exact fraction for T is recomputed by the verifier; its canonical numerator/denominator text is bound into the receipt by SHA-256.

5.3 Quantitative Newton lemma

We use the following standard form of the Newton–Kantorovich theorem.

Lemma 5.1

Let F be holomorphic on a convex neighbourhood of x_0 in $\mathbb{C}^n$. Suppose $DF(x_0)$ is invertible and

$$\|DF(x_0)^{-1}\| \leq \beta, \quad \|F(x_0)\| \leq \delta,$$

and suppose DF is Lipschitz with constant L on the ball of radius 2η , where $\eta = \beta\delta$. If

$$h = \beta L \eta \leq \frac{1}{2},$$

then F has a zero in that ball. The zero lies within

$$t_* = \frac{1 - \sqrt{1 - 2h}}{\beta L} \leq 2\eta$$

of x_0 . If $2\beta L \eta < 1$, its Jacobian is invertible.

Proof This is the Newton–Kantorovich majorant theorem applied to the scalar majorant $\beta\delta - t + (\beta L/2)t^2$. Its smaller root is t_* . The inequality $t_* \leq 2\eta$ follows from $1 - \sqrt{1 - 2h} \leq 2h$. Finally,

$$\|DF(x_0)^{-1}(DF(x) - DF(x_0))\| \leq 2\beta L \eta < 1$$

on the root enclosure, so Neumann inversion proves nonsingularity. $\square$

5.4 Cauchy and Newton constants

Equations (5.5)–(5.13) give, for every row and every $m \geq 5000$,

$$\epsilon_m := \sup_{(5.1)} |\Delta_{M,q}^{(m)}| \leq \frac{254}{m} + \frac{15374}{m^2} + \frac{1}{400000}, \quad (5.14)$$

and at the first-three-row centre,

$$\delta_m := \max |P_{M,q}^{(m)}(X_r^*)| \leq \frac{2}{m} + \frac{1105}{m^2} + \frac{1}{400000}. \quad (5.15)$$

At $m = 5000$,

$$\epsilon_m \leq 0.05141746, \quad \delta_m \leq 0.0004467. \quad (5.16)$$

The first-derivative Cauchy row-sum coefficient at the centre is 12, for which we use 15. On the inner ball of radius $1/100$, the distance to the boundary of (5.1) is $6/25$. The second-derivative Cauchy row-sum coefficient is

$$\frac{12}{(6/25)^2} = \frac{625}{3} < 300. \quad (5.17)$$

The limit Hessian row sum is at most $188.468 < 258$, and the fourth-row gradient is at most $27.095 < 37$. All four comparisons are derived and checked in the executable receipt.

Let $\beta_0 = 67/100$. Neumann inversion gives

$$\beta_m \leq \frac{\beta_0}{1 - 15\beta_0\epsilon_m}. \quad (5.18)$$

Let

$$L_m = 258 + 300\epsilon_m, \quad \eta_m = \beta_m\delta_m, \quad h_m = \beta_m L_m \eta_m. \quad (5.19)$$

At the endpoint $m = 5000$, exact rational evaluation gives

predicate	certified bound
$15\beta_0\epsilon_m$	$0.516745473 < 1$
β_m	1.386432952
L_m	273.425238
h_m	$0.234775241 < 1/2$
$2\eta_m$	$0.001238640 < 1/100$
fourth-row margin	$1.392752889 > 0$
transported Jacobian margin	$0.530449518 > 0$

The Newton–Kantorovich theorem therefore supplies a zero of the first three finite rows within $2\eta_m$ of X_r^* . The last two margins show that the fourth row and the first-three-row Jacobian remain nonzero at that zero. Every defect expression decreases with m , so the endpoint proves all $m \geq 5000$.

Theorem 5.2 (uniform analytic branch)

For every $m \geq 5000$ and $r \in \{0, 2, 3\}$, the first three rows associated with $d = 4m + r$ have a common zero at which their anchored Jacobian and the fourth row are nonzero.

6. Finite interval certificates

It remains to cover $5 \leq m < 5000$ in residues 0, 2, 3.

6.1 Exact coefficient evaluation and rigorous tails

For $m \leq 30$, the checker evaluates every coefficient in (3.3). For $m \geq 31$, it evaluates all monomials of total degree at most 50. Let $R = 4/5$, let $U = 51$, and put

$$E_{k,u} = \frac{(5m + 8 + u)^u (3R)^u u^k}{m^u u! R^k}, \quad k = 0, 1, 2. \quad (6.1)$$

This bounds respectively the omitted value, gradient row sum and Hessian row sum at level u . Support gives $u \leq d + 3 \leq 4m + 6$, so

$$\frac{u}{5m + 8 + u} < \frac{1}{2}.$$

Hence, with an Arb upper bound for $e^{1/2}$, the first omitted ratio is

$$q_{k,U} = \frac{3R(5m + 9 + U)e^{1/2}}{m(U + 1)} \left(\frac{U + 1}{U} \right)^k < 1. \quad (6.2)$$

The right side decreases at later levels, so the complete tail is bounded by

$$\frac{E_{k,U}}{1 - q_{k,U}}. \quad (6.3)$$

For completeness, after removing constants the u -dependent factors in the ratio majorant are respectively

$$\frac{5m + 9 + u}{u + 1}, \quad \frac{5m + 9 + u}{u}, \quad \frac{(5m + 9 + u)(u + 1)}{u^2}$$

for $k = 0, 1, 2$. Each is decreasing, which justifies using the first omitted ratio for the complete supported tail.

6.2 Interval-Newton gate

Proposed centres are exact rational decimal points. For $5 \leq m \leq 16$, they are stored explicitly; for $m \geq 17$, the checker uses one fixed formula

$$X_{m,r}^{\text{loc}} = X_r^* + \frac{Y_r}{m}, \quad (6.4)$$

where the decimal rational vectors Y_r are part of the hashed source. Their numerical origin is irrelevant to acceptance.

The checker evaluates each truncated row, all first derivatives and all second derivatives in 128-bit complex Arb arithmetic. The derivative box is the coordinate rectangle of real and imaginary radius $1/100$. Its maximum complex displacement is therefore $\sqrt{2}/100$, and the checker proves that the whole rectangle lies in $|X_s| < 4/5$, the domain used by (6.1).

Let J_0 be the truncated centre Jacobian and let $\beta_{\text{tr}} = \|J_0^{-1}\|_{\infty}$. If T_1 is the gradient-tail bound, Neumann inversion gives

$$\beta = \frac{\beta_{\text{tr}}}{1 - \beta_{\text{tr}}T_1}. \quad (6.5)$$

Let δ be the first-three-row residual including T_0 , and let L be the Hessian row-sum bound on the box including T_2 . Each case is accepted only if the explicit outward-rounded endpoints prove

$$h = \beta L(\beta\delta) < \frac{1}{2}, \quad 2\beta\delta < \frac{1}{100}, \quad (6.6)$$

and if interval transport proves both the fourth row and the Jacobian nonzero throughout the resulting root enclosure.

6.3 Coverage and worst case

The ledger contains exactly

$$4995 \times 3 = 14985$$

records, ordered lexicographically by $m = 5, \dots, 4999$ and $r = 0, 2, 3$. Thirty-six cases use stored centres and 14949 use (6.4). The independent structural auditor reports the following worst endpoints:

quantity	certified endpoint	case
maximum h	$0.465782636 < 1/2$	$m = 17, r = 2$
maximum root radius	$0.006955535 < 1/100$	$m = 17, r = 2$
minimum fourth-row margin	$1.475688755 > 0$	$m = 17, r = 2$
minimum Jacobian margin	$0.068434729 > 0$	$m = 17, r = 2$

The normal and optimized runs produce byte-identical ledgers with SHA-256

a106c2df4bf6b771b0ee3245d6e0f5c95d1a1cb5485e9b03e93d942f77ccc3a1

The independent audit receipt has SHA-256

d4702c869d6d735a3736ece0a19c61232e667b8dca6ba0c7277e1b5cd89923b6

Theorem 6.1 (finite bridge)

For every $5 \leq m < 5000$ and $r \in \{0, 2, 3\}$, the first three rows associated with $d = 4m + r$ have a common zero at which their anchored Jacobian and the fourth row are nonzero.

7. Exact residue $r = 1$

Let $d = 4m + 1$ and set $X = (0, 0, 0)$. Congruence excludes a constant term from rows $q = 1, 2, 3$, so the first three rows vanish. Their only linear monomials are respectively X_1, X_2, X_3 , and their Jacobian is diagonal:

$$\text{diag} \left(-\frac{5m+2}{m}, -\frac{5m+3}{m}, -\frac{5m+4}{m} \right). \quad (7.1)$$

Its determinant is nonzero for every $m \geq 1$. The wrapped fourth row has $q = 0, M = m + 1$ and constant term exactly 1. Thus this residue class is algebraic and needs no interval computation. This specialization recovers the $e = 4, d - 1 \equiv 0 \pmod{4}$ case of Edo's divisibility theorem [4].

8. Proof of Theorem A

Lewis–Perry–Straub prove (1.4) for $2 \leq d < 20$. For $d \geq 20$, write $d = 4m + r$, so $m \geq 5$. Section 7 proves the required specialization when $r = 1$. Theorem 6.1 proves it for $r = 0, 2, 3$ and $m < 5000$, while Theorem 5.2 proves it for $m \geq 5000$.

Unscaling gives

$$P = (X_1/m, X_2/m, X_3/m, 1).$$

All row-normalization factors $C_{M,q}$ are nonzero. Coordinate scaling and row normalization multiply the anchored Jacobian by nonzero factors, so the certified normalized determinant is nonzero exactly when the anchored minor in Proposition 2.1 is nonzero. Proposition 2.1 then supplies the full determinant condition in Published input 1.1. Applying the Lewis–Perry–Straub implication proves (1.4). $\square$

9. Reproducibility and trust boundary

The proof package pins Python 3.14.6, python-flint 0.8.0 and the Arb precision used by each checker. Its proof-critical parts are:

1. `e4_eventual_envelope.py`, which recomputes the exact head/tail bounds, the 256-bit Fourier enclosures and the threshold inequalities;
2. `e4_certify_prototype.py`, the coefficient/evaluation/interval-Newton core;
3. `e4_finite_bridge.py`, the ordered finite producer;
4. `verify_finite_ledger.py`, the separate structural and endpoint auditor;
5. `verify_normalization.py`, which independently reconstructs 78,144 admissible normalized coefficients from the multinomial definition and checks 1,446,940 rejected support triples;
6. the exact locator file; and
7. the normal and optimized finite ledgers and receipts.

The proof-critical frozen digests are listed in four 16-character groups; spaces are typographical only:

- `eventual` receipt: 747cd56f4a297774 0ac0786055d2cdf0 7ed9e7b89521ed91
0ad736d6eaf5daa9;
- `normalization` receipt: b82f1709aaaae6e1f cc58a8091125c11d cb3ff5b0b9095fc9
85c2bf516a62d47e;

- finite v2 ledger: a106c2df4bf6b771 b0ee3245d6e0f5c9 5d1a1cb5485e9b03 e93d942f77ccc3a1;
- finite v2 production receipt: 4e14868e1531fb85 509dc017533b9261 21c130d31ef1eb1b 41195d5cfddb5b54;
- independent ledger audit: d4702c869d6d735a 3736ece0a19c6123 2e667b8dca6ba0c7 277e1b5cd89923b6.

The decimal locators are not trusted as solutions. The finite proof depends on the interval gates in (6.6), exact coefficient formulas, and rigorous omitted tails. Likewise, the finite ledger is not independent reproduction: both runs use the same source and arithmetic library. Its evidential role is exhaustive replay and optimization-sensitivity detection.

Arb midpoint-radius arithmetic is described in [3]. A specialist verifier can replace the implementation while preserving the mathematical certificate conditions in Sections 5 and 6.

10. Limitations and claim boundary

- The finite range is a computer-assisted proof, not a hand enumeration.
- The implication from the coefficient specialization to the stratum containment is the published theorem of Lewis–Perry–Straub; it is not re-formalized in a proof assistant here.
- The package has been internally replayed, but not independently reproduced or externally peer reviewed.
- The search found no earlier all-degree $e = 4$ proof, but this does not establish priority.
- The theorem concerns plane polynomial automorphism strata. It neither proves nor refutes the two-dimensional Jacobian conjecture, the quartic Hessian conjecture in dimension four, or any other open Jacobian/Hessian case.

References

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Declarations

Data and code availability. The exact frozen successor archive, manifest, source files, ledgers and receipts accompany the candidate release.

Author responsibility. The submitting author is responsible for the mathematical claims, computational evidence and source selection.

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